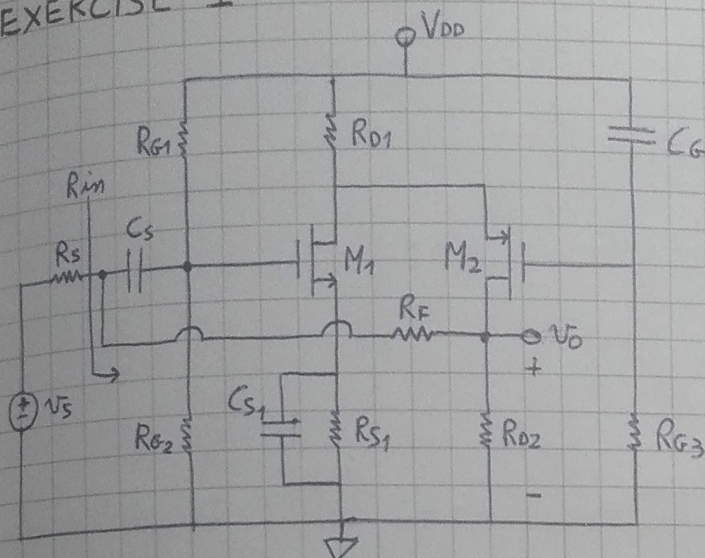


EXERCISE 1



$$V_{DD} = 10V$$

$$R_{G1} = 150k\Omega$$

$$R_{G2} = 100k\Omega$$

$$R_{G3} = 100k\Omega$$

$$R_{S1} = 1k\Omega$$

$$R_{D2} = 15k\Omega$$

$$R_F = 15k\Omega$$

$$R_S = 1k\Omega$$

$$C_S = 10\mu F$$

$$C_G = 10\mu F$$

$$C_{S1} = 10\mu F$$

M1 (nMOS):

$$V_{t1} = 2V$$

$$I_{DSS1} = 2.4mA$$

$$\mu_0 = +\infty$$

M2 (pMOS):

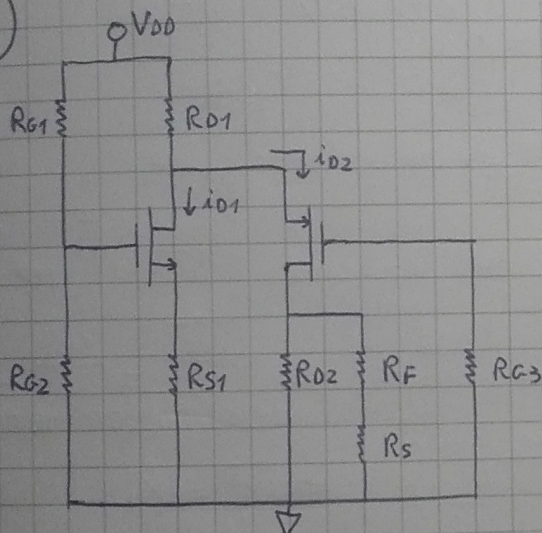
$$V_{t2} = -3V$$

$$I_{DSS2} = 3.6mA$$

$$\mu_0 = +\infty$$

1) FIND THE VALUE OF R_{D1} SUCH THAT $V_O = V_{D2} = 2V$

DC)



$$V_{G1} = V_{DD} \frac{R_{G2}}{R_{G1} + R_{G2}} = 4V$$

$$V_{GS1} = V_{G1} - V_{S1} = V_{G1} - i_{D1} R_{S1}$$

$$i_{D1} = I_{DSS1} \left(\frac{V_{GS1}}{V_{t1}} - 1 \right)^2$$

METHOD 1: $i_{D1} = I_{DSS1} \left(\frac{V_{G1} - i_{D1} R_{S1}}{V_{t1}} - 1 \right)^2$ SOLVE PARABOLA WITH i_{D1} AS VARIABLE

METHOD 2: $i_{D1} = \frac{V_{G1} - V_{GS1}}{R_{S1}} \Rightarrow \frac{V_{G1} - V_{GS1}}{R_{S1}} = I_{DSS1} \left(\frac{V_{GS1}}{V_{t1}} - 1 \right)^2$ SOLVE PARABOLA WITH V_{GS1} AS VARIABLE

It's strongly recommended to solve the parabola with V_{GS1} as variable, the calculations are much easier and they take less time.

$$\frac{V_{G1}}{R_{S1}} - \frac{V_{GS1}}{R_{S1}} = i_{DSS1} \left(1 + \frac{V_{GS1}^2}{V_{t1}^2} - 2 \frac{V_{GS1}}{V_{t1}} \right) \Rightarrow \underbrace{\left(\frac{i_{DSS1}}{R_{S1}} \right)}_A V_{GS1}^2 + \underbrace{\left(\frac{1}{R_{S1}} - 2 \frac{i_{DSS1}}{V_{t1}} \right)}_B V_{GS1} + \underbrace{\left(\frac{i_{DSS1}}{R_{S1}} - \frac{V_{G1}}{R_{S1}} \right)}_C = 0$$

$$A = 0,6 \cdot 10^{-3}$$

$$B = -1,4 \cdot 10^{-3}$$

$$C = -1,6 \cdot 10^{-3}$$

$$V_{GS1} = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} = \begin{cases} 3,17V & \text{OK} \\ -0,83V & \text{NOT OK} \end{cases}$$

$V_{GS} > 0$ for an nMOS in saturation regime

$$V_{S1} = V_{G1} - V_{GS1} = 0,83V$$

$$i_{D1} = \frac{V_{S1}}{R_{S1}} = 830 \mu A$$

$$i_{D2} = \frac{V_{D2}}{R_{D2} \parallel (R_F + R_S)} = 258 \mu A$$

METHOD 1: $i_{D2} = i_{DSS2} \left(\frac{V_{GS2}}{V_{t2}} - 1 \right)^2 \Rightarrow \frac{i_{D2}}{i_{DSS2}} = \left(\frac{V_{GS2}}{V_{t2}} - 1 \right)^2$ SOLVE PARABOLA

$$\Rightarrow \left(\frac{1}{V_{t2}} \right) V_{GS2}^2 + \left(-2 \frac{1}{V_{t2}} \right) V_{GS2} + \left(1 - \frac{i_{D2}}{i_{DSS2}} \right) = 0$$

$$A = 0,111$$

$$B = 0,667$$

$$C = 0,928$$

$$V_{GS2} = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} = \begin{cases} -2,19V & \text{NOT OK} \\ -3,815V & \text{OK} \end{cases}$$

$V_{GS} < V_t$ for a pMOS in saturation regime

METHOD 2: Notice i_{D2} is a known value, so $i_{D2} = i_{DSS2} \left(\frac{V_{GS2}}{V_{t2}} - 1 \right)^2 \Rightarrow$

$$\Rightarrow \frac{i_{D2}}{i_{DSS2}} = \left(\frac{V_{GS2}}{V_{t2}} - 1 \right)^2 \Rightarrow \sqrt{\frac{i_{D2}}{i_{DSS2}}} = \left| \frac{V_{GS2}}{V_{t2}} - 1 \right| \Rightarrow \sqrt{\frac{i_{D2}}{i_{DSS2}}} = \frac{V_{GS2}}{V_{t2}} - 1$$

MUCH FASTER METHOD

Little algebra recap: remember that for $x \in \mathbb{R} \Rightarrow \sqrt{x^2} = |x|$.

There are two versions of the i_D formula: some people use $i_D = i_{DSS} \left(\frac{V_{GS}}{V_t} - 1 \right)^2$, others use $i_D = i_{DSS} \left(1 - \frac{V_{GS}}{V_t} \right)^2$. The two formulas are equivalent, but remember that for a MOS in saturation $|V_{GS}| > |V_t|$ (in this case, having a pMOS, $V_{GS} < V_t < 0$). So if you use the first formula $\left| \frac{V_{GS}}{V_t} - 1 \right| = \frac{V_{GS}}{V_t} - 1$, if you use the second one $\left| 1 - \frac{V_{GS}}{V_t} \right| = -1 + \frac{V_{GS}}{V_t}$.

$$\Rightarrow V_{GS2} = V_{t2} \left(\sqrt{\frac{i_{D2}}{i_{DSS2}}} + 1 \right) = -3,82V$$

$$V_{G2} = 0 \Rightarrow V_{S2} = V_{G2} - V_{GS2} = -V_{GS2} = 3,82V$$

$$R_{D1} = \frac{V_{DD} - V_{S2}}{i_{D1} + i_{D2}} = 5,68 k\Omega$$

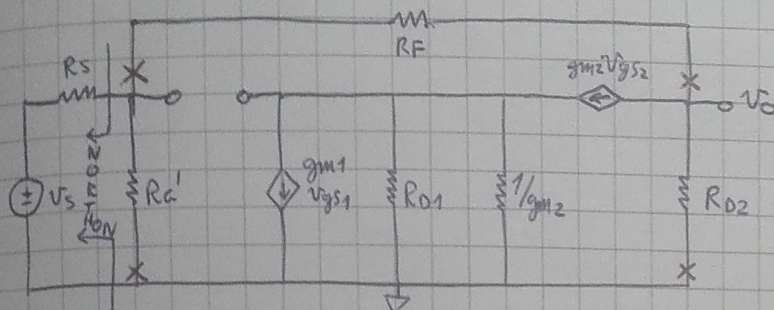
2) FIND THE OPERATING POINTS (V_{DS}, I_{DS}) OF M_1 AND M_2

$$V_{D1} = V_{S2} = 3,82 \text{ V} \Rightarrow V_{DS1} = V_{D1} - V_{S1} = 2,99 \text{ V}$$

$$V_{DS2} = V_{D2} - V_{S2} = -1,82 \text{ V}$$

3) FIND THE VOLTAGE GAIN OF THE AMPLIFIER A_v

AC)

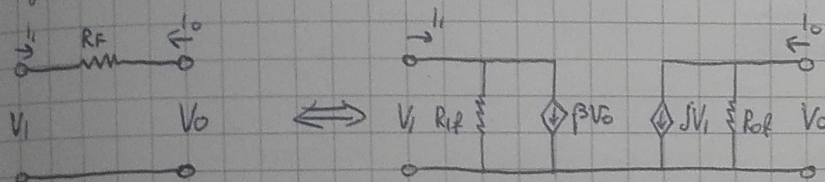


IT'S A VOLTAGE SAMPLING
CURRENT MIXING

CURRENT MIXING $\Rightarrow$ NORTON ON INPUT
GENERATOR

$$R_G' = R_{G1} // R_{G2} = 60 \text{ k}\Omega$$

$1/g_{m2}$ = INPUT RESISTANCE OF COMMON GATE STAGE



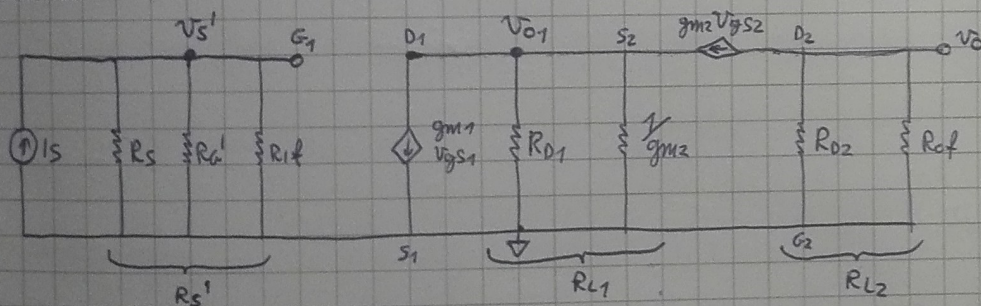
$$R_{if} = \frac{V_i}{I_i} \bigg|_{V_o=0} = R_F = 15 \text{ k}\Omega$$

$$R_{of} = \frac{V_o}{I_o} \bigg|_{V_i=0} = R_F = 15 \text{ k}\Omega$$

$$\beta = \frac{I_i}{V_o} \bigg|_{V_i=0} = -\frac{1}{R_F} = -66,7 \cdot 10^{-6} \text{ S}$$

$\beta \rightarrow 0$

OPEN LOOP!



$$I_S = \frac{v_s}{R_S}$$

In this case I can't write v_{o1} in function of the only v_{gs1} or v_{gs2} because this time I can't separate the two variables. I have to use the single stage amplifications.

$$M1: \text{COMMON SOURCE} \Rightarrow A_{v1} = \frac{-g_{m1} R_{L1}}{1 + g_{m1} R_{S1}} = -g_{m1} R_{L1} \quad \text{BECAUSE } R_{S1} = 0$$

$$M2: \text{COMMON GATE} \Rightarrow A_{v2} = g_{m2} R_{L2}$$

$$g_{m1} = \frac{2}{V_{t1}} \sqrt{I_{D1} I_{DSS1}} = 1,41 \text{ mS}$$

$$g_{m2} = \frac{2}{|V_{t2}|} \sqrt{I_{D2} I_{DSS2}} = 642 \mu\text{S}$$

$$R_S' = R_S // R_G' // R_{if} = 923 \Omega$$

$$R_{L1} = R_{O1} // \frac{1}{g_{m2}} = 1,22 \text{ K}\Omega$$

$$R_{L2} = R_{O2} // R_{O2} = 7,5 \text{ K}\Omega$$

$$\begin{cases} v_{S'} = i_S R_S' \\ \frac{v_O}{v_{S'}} = A_{v1} \cdot A_{v2} \end{cases} \Rightarrow A_{OL} = \frac{v_O}{i_S} = R_S' A_{v1} A_{v2}$$

$$A_{v1} = -1,72$$

$$A_{v2} = 4,82$$

$$\Rightarrow A_{OL} = -7,64 \text{ K}\Omega$$

CLOSED LOOP:

$$A_F = \frac{A_{OL}}{1 + \beta A_{OL}} = -5,06 \text{ K}\Omega = \frac{v_O}{i_S}$$

$$A_v = \frac{v_O}{v_S} = A_F \cdot \frac{1}{R_S} = -5,06$$

4) FIND THE INPUT RESISTANCE R_{in} INDICATED IN THE FIGURE

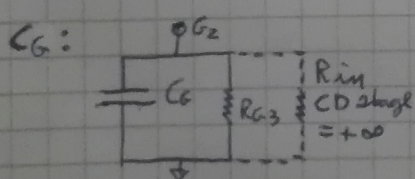
R_S is not included in the calculation of R_{in} , but due to the interaction with the feedback R_S must be removed after the calculation of R_{in}^F and not before.

$$R_{in}^d = R_S'$$

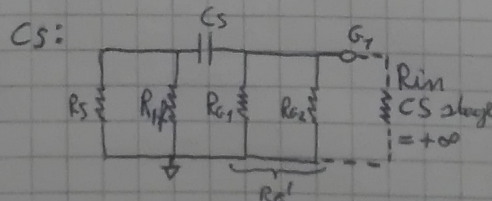
$$R_{in}^F = \frac{R_{in}^d}{1 + \beta A_{OL}} = 619 \Omega$$

$$R_{in} = \frac{1}{\frac{1}{R_{in}^F} - \frac{1}{R_S}} = 1,63 \text{ K}\Omega \quad \text{PARALLEL REMOVING OF A RESISTANCE}$$

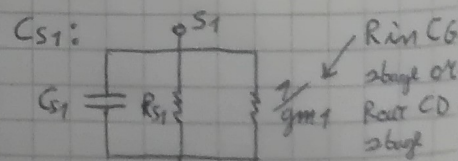
5) FIND THE LOW FREQUENCY BANDWIDTH OF THE AMPLIFIER



$$R_{CG} = R_{G3} = 100 \text{ K}\Omega$$



$$R_{CS} = R_S // R_{if} + R_G' = 60,9 \text{ K}\Omega$$



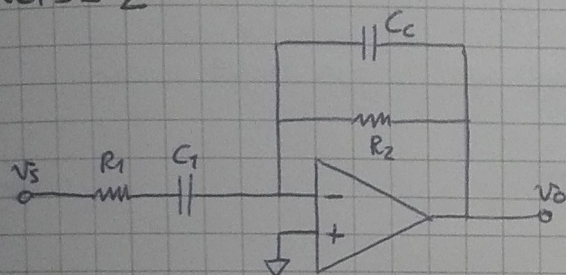
$$R_{CS1} = R_{S1} // \frac{1}{g_{m1}} = 475 \Omega$$

$$f_L^d = \frac{1}{2\pi} \left(\frac{1}{R_{CG} \cdot C_G} + \frac{1}{R_{CS} \cdot C_S} + \frac{1}{R_{CS1} \cdot C_{S1}} \right) = 38,8 \text{ Hz}$$

$$f_L^F = \frac{f_L^d}{1 + \beta A_{OL}} = 25,7 \text{ Hz}$$

NOTE: While studying the resistance seen from the capacitor C_0 , the transistor M_2 becomes configured in Common Drain stage, having the input on the gate (where there is the capacitor C_0) and the output on the source. It could also be considered a Common Source stage, the resulting R_{in} doesn't change ($R_{inCS} = R_{inCD} = \infty$). The same happens for C_{S1} capacitor. For capacitor C_{S1} we can consider the transistor M_1 in Common Gate stage (with $R_{inCG} = 1/g_{m1}$) or in Common Drain stage (with $R_{inCD} = 1/g_{m1}$). Also in this case the resistance we're interested in doesn't change.

EXERCISE 2



$$A(s) = \frac{A_{d0}}{(1 + \frac{s}{\omega_0})(1 + \frac{s}{\omega_1})}$$

$$\begin{aligned} R_1 &= 10 \text{ k}\Omega \\ C_1 &= 100 \text{ nF} \\ R_2 &= 220 \text{ k}\Omega \end{aligned}$$

$$\begin{aligned} A_{d0} &= 10^5 \text{ V/V} \\ \omega_0 &= 10 \text{ rad/s} \\ \omega_1 &= 5000 \text{ rad/s} \end{aligned}$$

NEGLECT THE CAPACITOR C_c . FIND:

1) THE EXPRESSION OF THE IDEAL CIRCUIT VOLTAGE GAIN $A_V(s)$

$$Z_1 = R_1 + \frac{1}{C_1 s} = \frac{R_1 C_1 s + 1}{C_1 s}, \quad Z_2 = R_2$$

this is an inverting amplifier, so the ideal voltage gain is:

$$A_V = -\frac{Z_2}{Z_1} = -\frac{R_2 C_1 s}{R_1 C_1 s + 1}$$

2) AN ESTIMATION OF THE PHASE MARGIN PM

$$\beta := \frac{v_-}{v_o} \bigg|_{v_s=0} = \frac{Z_1}{Z_1 + R_2} = \frac{R_1 C_1 s + 1}{(R_1 + R_2) C_1 s + 1}$$

$$\Rightarrow \beta_0 = 1$$

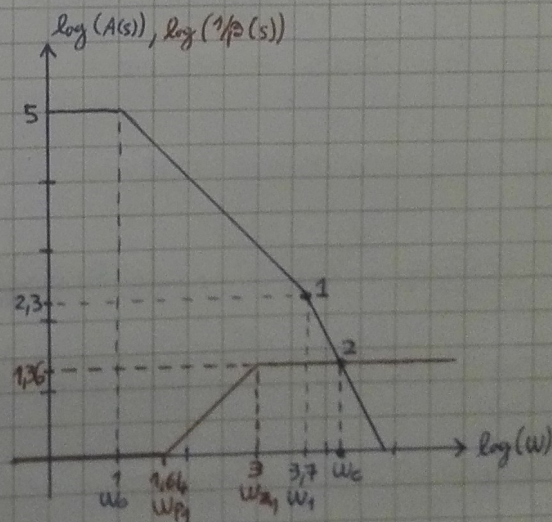
$$\Rightarrow \frac{1}{\beta_0} = 1$$

$$\beta_\infty = \frac{R_1}{R_1 + R_2} = \frac{1}{23}$$

$$\frac{1}{\beta_\infty} = 23$$

$$\omega_{z1} = \frac{1}{R_1 C_1} = 1 \text{ k rad/s}$$

$$\omega_{p1} = \frac{1}{(R_1 + R_2) C_1} = 43,5 \text{ rad/s}$$

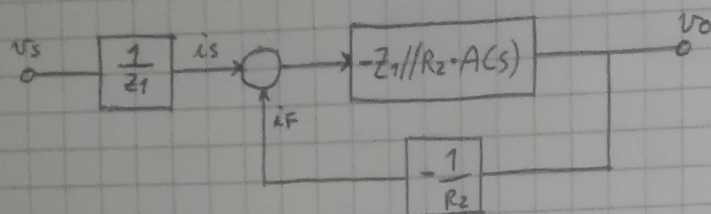


$$\frac{A_1}{A_2} = \left(\frac{\omega_2}{\omega_1}\right)^2 \Rightarrow \frac{10^{23}}{10^{1.3}} = \left(\frac{\omega_c}{10^{3.7}}\right)^2 \Rightarrow \omega_c = 14,74 \text{ rad/s}$$

$$PM = 180 - \arctan\left(\frac{\omega_c}{\omega_0}\right) - \arctan\left(\frac{\omega_c}{\omega_1}\right) - \arctan\left(\frac{\omega_c}{\omega_{p1}}\right) + \arctan\left(\frac{\omega_c}{\omega_{z1}}\right) = 75,06^\circ$$

3) THE BLOCK DIAGRAM OF THE AMPLIFIER, ACCORDING TO THE FEEDBACK THEORY

The inverting amplifier is a voltage sampling - current mixing, so we need to visualize the involved currents in the circuit.



CURRENT MIXING $\Rightarrow$ COMPARISON OF CURRENTS

4) THE EXPRESSION OF THE INPUT SCALING FUNCTION $u_s(s)$ IN THE ABOVE DIAGRAM

$$u_s(s) = \frac{1}{z_1} = \frac{C_1 s}{R_1 C_1 s + 1}$$

NOW WE TAKE IN ACCOUNT C_c . FIND:

5) THE MAXIMUM VALUE OF THE CAPACITOR C_c FOR WHICH THE AMPLIFIER

MANTAINS THE SAME FREQUENCY RESPONSE UP TO THE CROSSOVER FREQUENCY ω_c

The addition of C_c introduces a zero whose frequency is:

$$\omega_{z2} = \frac{1}{R_2 C_c}$$

If we want to maintain the same frequency response up to the crossover frequency

ω_z must be greater or equal to ω_c . Due to the fact we need to find the maximum value of C_c which allows us to do this, we need $\omega_z = \omega_c$

$$\Rightarrow C_c = \frac{1}{R_2 \omega_{z2}} = 308 \text{ pF}$$

6) THE NEW PHASE MARGIN OF THE CIRCUIT

$$p(s) = \frac{z_1}{z_1 + z_2}, \quad z_2 = R_2 \parallel \frac{1}{C_2 s} = \frac{R_2}{R_2 C_2 s + 1}$$

$$\Rightarrow \beta(s) = \frac{R_1 R_2 C_1 C_2 s^2 + (R_1 C_1 + R_2 C_2) s + 1}{R_1 R_2 C_1 C_2 s^2 + ((R_1 + R_2) C_1 + R_2 C_2) s + 1}$$

$\rightarrow$ ACTUALLY IT DOES, BUT THE CHANGES ARE NEGLIGIBLE

IMPORTANT! The introduction of C_c doesn't change the values of the previous poles and zeros found. This means that I already have the values of one pole and two zeros, so I only have to find the value of the last pole.

o METHOD 1: SOLVE PARABOLA

I don't need to solve the parabola at the numerator because I already have the values of the two zeros. I just need to solve the one at the denominator.

$$A = R_1 R_2 C_1 C_2 = 67,76 \cdot 10^{-9}$$

$$B = (R_1 + R_2) C_1 + R_2 C_2 = 23,07 \cdot 10^{-3}$$

$$C = 1$$

$$\omega_{p12} = \begin{cases} 43,5 \frac{\text{rad}}{\text{s}} \\ 339 \text{ K} \frac{\text{rad}}{\text{s}} \end{cases} \text{ OK, IT'S THE POLE I ALREADY HAVE}$$

o METHOD 2: I notice that $\beta_{\infty} = 1$ (and, as a consequence, $\frac{1}{\beta_{\infty}} = 1$) due to the short circuit C_2 makes on R_2 at infinite frequencies. So if $\frac{1}{\beta(\omega_{z2})} = 23$ before starting decreasing with -20 dB/dec , then ω_{p2} must be a frequency 23 times bigger than ω_{z2} . See the drawing below.

$$\omega_{p2} = 23 \cdot \omega_{z2} = 339 \text{ K} \frac{\text{rad}}{\text{s}} \quad \text{I CAN COMPLETELY AVOID THE CALCULATION OF } \beta(s)$$

$$PM = 15,06 - \arctan\left(\frac{\omega_c}{\omega_{p2}}\right) + \arctan\left(\frac{\omega_c}{\omega_{z2}}\right) = 57,57^\circ$$

$\uparrow$ PM FOUND BEFORE $\uparrow$ = 1

